

Copyright © 2017, Ed Skoudis and John Strand. All rights reserved to Ed Skoudis and John Strand and/or SANS Institute.

PLEASE READ THE TERMS AND CONDITIONS OF THIS COURSEWARE LICENSE AGREEMENT ("CLA") CAREFULLY BEFORE USING ANY OF THE COURSEWARE ASSOCIATED WITH THE SANS COURSE. THIS IS A LEGAL AND ENFORCEABLE CONTRACT BETWEEN YOU (THE "USER") AND THE SANS INSTITUTE FOR THE COURSEWARE. YOU AGREE THAT THIS AGREEMENT IS ENFORCEABLE LIKE ANY WRITTEN NEGOTIATED AGREEMENT SIGNED BY YOU.

With the CLA, the SANS Institute hereby grants User a personal, non-exclusive license to use the Courseware subject to the terms of this agreement. Courseware includes all printed materials, including course books and lab workbooks, as well as any digital or other media, virtual machines, and/or data sets distributed by the SANS Institute to the User for use in the SANS class associated with the Courseware. User agrees that the CLA is the complete and exclusive statement of agreement between The SANS Institute and you and that this CLA supersedes any oral or written proposal, agreement or other communication relating to the subject matter of this CLA.

BY ACCEPTING THIS COURSEWARE, YOU AGREE TO BE BOUND BY THE TERMS OF THIS CLA. BY ACCEPTING THIS SOFTWARE, YOU AGREE THAT ANY BREACH OF THE TERMS OF THIS CLA MAY CAUSE IRREPARABLE HARM AND SIGNIFICANT INJURY TO THE SANS INSTITUTE, AND THAT THE SANS INSTITUTE MAY ENFORCE THESE PROVISIONS BY INJUNCTION (WITHOUT THE NECESSITY OF POSTING BOND), SPECIFIC PERFORMANCE, OR OTHER EQUITABLE RELIEF.

If you do not agree, you may return the Courseware to the SANS Institute for a full refund, if applicable.

User may not copy, reproduce, re-publish, distribute, display, modify or create derivative works based upon all or any portion of the Courseware, in any medium whether printed, electronic or otherwise, for any purpose, without the express prior written consent of the SANS Institute. Additionally, User may not sell, rent, lease, trade, or otherwise transfer the Courseware in any way, shape, or form without the express written consent of the SANS Institute.

If any provision of this CLA is declared unenforceable in any jurisdiction, then such provision shall be deemed to be severable from this CLA and shall not affect the remainder thereof. An amendment or addendum to this CLA may accompany this courseware.

SANS acknowledges that any and all software and/or tools, graphics, images, tables, charts or graphs presented in this courseware are the sole property of their respective trademark/registered/copyright owners, including:

AirDrop, AirPort, AirPort Time Capsule, Apple, Apple Remote Desktop, Apple TV, App Nap, Back to My Mac, Boot Camp, Cocoa, FaceTime, FileVault, Finder, FireWire, FireWire logo, iCal, iChat, iLife, iMac, iMessage, iPad, iPad Air, iPad Mini, iPhone, iPhoto, iPod, iPod classic, iPod shuffle, iPod nano, iPod touch, iTunes, iTunes logo, iWork, Keychain, Keynote, Mac, Mac Logo, MacBook, MacBook Air, MacBook Pro, Macintosh, Mac OS, Mac Pro, Numbers, OS X, Pages, Passbook, Retina, Safari, Siri, Spaces, Spotlight, There's an app for that, Time Capsule, Time Machine, Touch ID, Xcode, Xserve, App Store, and iCloud are registered trademarks of Apple Inc.

Governing Law: This Agreement shall be governed by the laws of the State of Maryland, USA.

Table of Contents		Page
Getting Networked		03
Workshop Objectives		10
Keeping Track of Your Work		13
Tools to Use Today		16
Ground Rules		17
Step 1: Reconnaissance		22
Step 2: Scanning		30
Step 3: Exploiting Systems		34
- Compromising Additional Machines		36
Step 4: Keeping Access		37
Step 5: Covering the Tracks		39
Building a Lab at Home		41
Capture the Flag Rules		43
Let the Games Begin!!		46
Appendix: Helpful Commands		50



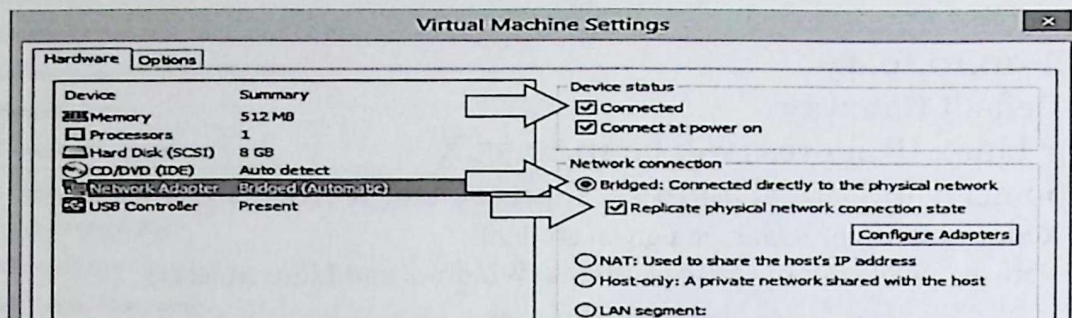
This table of contents can be used for future reference.

THE MOST IMPORTANT SECTION IN TODAY'S BOOK IS THE ONE SPELLING OUT THE GROUND RULES. MAKE SURE YOU FOLLOW THE GROUND RULES!

Ignoring the ground rules could subject you to ejection from the class and the conference! That would be very, very unfortunate, so do yourself a favor: FOLLOW THESE SIMPLE AND REASONABLE GROUND RULES!

VMware Bridged Networking

- In VMware, use bridged networking
 - Host-only doesn't make sense
 - NAT will get in your way
- Press CTRL-D, and select "Network Adapter"



Configure both your Windows and Linux guest machines to use bridged networking. In VMware (Workstation or Player), go to VM→Settings. Click "Network Adapter," and select "Bridged" networking. Select "Replicate physical network connection state." Then, make sure that you have a check next to "Connected" and "Connect at power on."

Setting Up Windows

- Access your network setup
 - Access your network interfaces by running `ncpa.cpl`
`C:\> ncpa.cpl`
 - Choose your Local Area Connection
 - Select TCP/IPv4 and click "Properties"
 - Set this to your given IP address: 10.10.76.X
 - Subnet mask should be 255.255.0.0
 - Preferred DNS Server is 10.10.10.45
 - No default gateway (leave it blank)

You should know how to configure your Windows network. Just in case you don't, here are some hints.

Target Information

- Host Name: _____
- Host IP Addr: _____
- Host OS: _____
- Host Version: _____
- Services: _____
- Notes: _____

- Host Name: _____
- Host IP Addr: _____
- Host OS: _____
- Host Version: _____
- Services: _____
- Notes: _____

- Host Name: _____
- Host IP Addr: _____
- Host OS: _____
- Host Version: _____
- Services: _____
- Notes: _____

- Host Name: _____
- Host IP Addr: _____
- Host OS: _____
- Host Version: _____
- Services: _____
- Notes: _____

You will fill out these sheets with the information you discover during your scans and attacks.

Most Important Tools to Use Today

- Very Important:
 - Nmap: Port scanning and OS fingerprinting (Linux)
 - Nessus: Vulnerability scanning (Linux)
 - Netcat: Backdoors and file transfer (Windows and Linux)
 - Enum: Determining users and groups, and password guessing (Windows)
 - Metasploit: Exploiting vulnerable targets (Linux and Windows)
 - John the Ripper: Password cracking (Linux and Windows)
 - Fgdump: Remote SAM password hash dumper (Windows)
- Everything you need to win the game is included in this environment

You should attempt to use all of the important tools (on your own or on a partner's machine). The others are nice to use, but they are not essential.

Your Turn – Part II

- Now that we have covered a simulated reconnaissance phase, it will soon be your turn to use the tools
- When the instructor gives you permission to attack, you will start from this slide
- You might want to bookmark this page so you can easily come back

You might want to dog-ear or bookmark this page because it is the place where you will return after the lecture component of this class.

Key Steps to Exploiting a System

1. Reconnaissance
2. Scanning
3. Exploiting Systems
 - Gaining Access
4. Keeping Access – Backdoors and Trojans
5. Covering the Tracks

Now that scanning is complete, you can attempt to gain access.

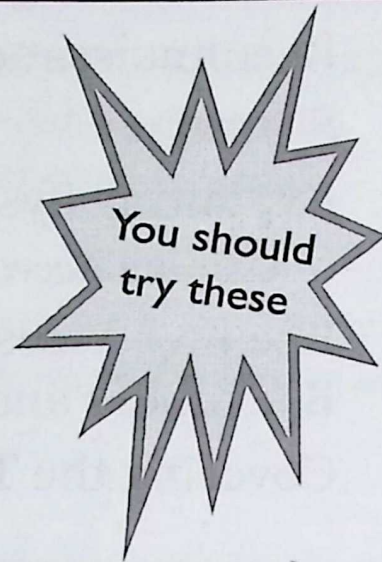
Compromising Additional Machines

- Once one machine is compromised, attackers can use it as a jumping off point for other attacks
 - Exploit Windows SMB sessions between target machines
 - Net use, at, etc.
 - Crack passwords, and look for systems where users have set up identical passwords on multiple machines

Remember, sometimes the easiest way in is through a full-frontal assault on a machine. For other systems, the easiest way in is to use information you've plundered from other boxes, exploiting trust relationships or shared accounts/passwords on the boxes.

Step 4: Keeping Access

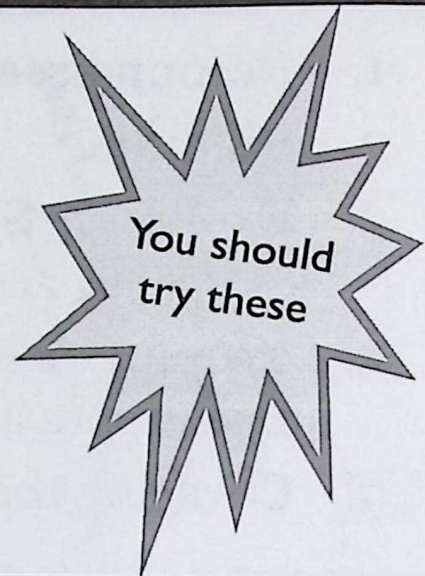
- Planting Netcat backdoor
- Use Metasploit shell or Meterpreter payloads
- Deploying VNC
- Others?
- DO NOT put Rootkits on the target machines; too risky



Feel free to deploy application-layer Trojan Horse backdoors, like Netcat or VNC. Or, use Metasploit to exploit a target, gaining remote shell or Meterpreter access of it.

Step 5: Covering the Tracks

- Creating hidden files on Linux
 - Directories named " "
- Creating hidden files on Windows
 - Alternate data streams
- Don't forget about shell history files!
 - Could be useful for you to see what others are attempting
 - You might want to cover your tracks by deleting your own shell histories on my machines



Once you've broken in, feel free to experiment with the hiding techniques we discussed in class.

In particular, hide your tracks on Linux and Windows by concealing files.

Capture the Flag Contest

- We'll play a game of capture the flag
- There are four regular flags and one bonus flag
 - flag1.txt, flag2.txt, flag3.txt, and flag4.txt
 - ... and bonusflag.txt
 - All flags located in the top of the directory structure (inside c:\ on Windows and / in Linux)
- Each flag provides you information about a "Phrase that pays"
- Break in to my machines, look at the flags, and determine the phrase that pays

We'll play a game of capture the flag. I have four flags on my machines, plus one bonus flag. You get to hack in and read the flags to determine the phrase that pays. The first one to determine the phrase that pays will win!

Are You Ready?

- Remember the attack process:
 1. Reconnaissance
 2. Scanning
 3. Exploiting Systems
 4. Keeping Access – Backdoors and Trojans
 5. Covering the Tracks
- Are there any questions on the ground rules or the Capture the Flag game?
- ASK NOW!!

Remember, follow these steps and the guidelines covered in this book. Are you ready?

LET THE GAMES BEGIN!! You Now Have Permission

- You now have permission to attack my systems on the 10.10.10.x network in this room
- Anything on the 10.10.10.x network in this room is a valid target...
- ...just follow the ground rules!
- Ask the instructor or a proctor for help and hints
 - We're happy to give you hints, but won't tell you exactly how to succeed
- Social engineering of the system administrator is also acceptable, but it better be good! 😊
 - E-mail, help desk calls ... no physical attacks, please



Okay, let us begin.

Remember, ask the instructor and/or proctor for help if you get stuck. We are here to give you hints. Also, I'll act as the system administrator for the target machines. My job involves keeping them up in light of the many attacks we'll see today. Every now and then, the instructor might have to reboot a box ... we'll work hard to keep that to a minimum, but it will occur.

Also, keep in mind that social engineering of the system administrator is also acceptable, but it better be good. Come see me if you think you have a clever social engineering exploit attempt.

Nmap

- Run an "Aggressive" Nmap scan (scan, OS fingerprint, version scan and NSE scripts) and save output to a file for future reference
 - # `nmap -A <target> --reason -o <file>`
- Scan specific port(s) on target
 - # `nmap -p <port(s)> <target> --reason`
- Perform a version scan on specific port(s)
 - # `nmap -sV -p <port(s)> <target> --reason`
- Additional options you might find helpful
 - `--reason` shows target response
 - `--packet_trace` shows packet details
 - `--traceroute` shows network topology

This page intentionally left blank.

Appendix: Helpful commands Pwdump

- To dump the passwords from a remote machine that you have an admin level user ID and password for
 - C:\> `pwdump3 10.10.10.9 [outfile] [user]`
- Then enter the password for the user id you used

This page intentionally left blank.

John The Ripper

- Linux: To unshadow a passwd file
 - # `unshadow /etc/passwd /etc/shadow > /tmp/combined`
- Linux: To crack an un-shadowed password file
 - # `john /tmp/combined`
- Windows: To crack a file with Windows hashes
 - `C:\> john <hash file>`
- Remember to delete "john.pot" when you want to restart a cracking session or it will pick up where it left off

This page intentionally left blank.

Remote Access

- VNC
 - \$ `vncviewer`
- SSH
 - \$ `ssh User@<TargetIP>`
- Telnet
 - \$ `telnet <TargetIP>`

This page intentionally left blank.

Miscellaneous

- To compile and run exploit code
 - \$ `gcc <exploit source> -o <outfile>`
 - \$ `./<outfile>`
- What user am I in Linux?
 - \$ `whoami`
 - \$ `id`
- Become root if you have the password
 - \$ `su -`

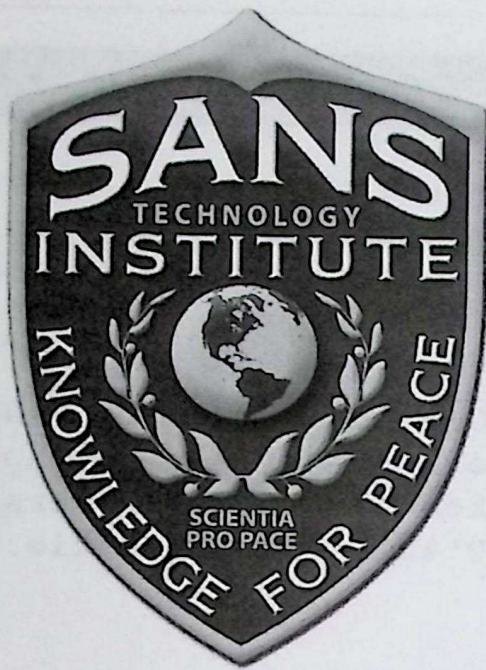
This page intentionally left blank.

Hydan

- To hide data
 - # `echo "Hello there." > hideme.txt`
 - # `./hydan ./ls hideme.txt > <outfile>`
- To retrieve data
 - # `./hydan-decode <stegofile>`
- Enter password when prompted

Hydan is in:

`/home/tools/hydan`



This Course is Part of the SANS Technology Institute (STI) Master's Degree Curriculum.

If your brain is hurting from all you've learned in this class, but you still want more, consider applying for a Master's Degree from STI. We offer two hands-on, intensive Master's Degree programs:

- Master of Science in Information Security Engineering
- Master of Science in Information Security Management

If you have a bachelor's degree and are ready to pursue a graduate degree in information security, please visit www.sans.edu for more information.

www.sans.edu

855-672-6733

info@sans.edu

Add N Edit Cookies	4:144
Address Resolution Protocol (ARP)	1:44, 1:252, 2:93, 2:142, 3:2, 3:47-49, 3:51-52, 3:56, 3:70-77, 3:79, 3:84-85, 3:93, 4:24, 4:175, 4:185-186, 6:17
Address Space Layout Randomization (ASLR)	5:56
Adobe Reader	3:114, 3:133
Adore	5:65
Advanced Intrusion Detection Environment (AIDE)	1:258, 5:67
Aggressive Mode	4:26
AIDE	1:258, 5:67
Airbase-NG	2:68
AirDefense	2:75
AirMagnet	2:75
Airopeek	2:64
AntiVirus (AV)	1:20, 1:53, 1:57, 1:119, 1:134-135, 1:164, 1:187, 1:193, 3:11, 3:29, 3:133, 3:136-138, 3:143-146, 4:29, 4:52, 4:62-63, 4:65, 4:72, 4:74, 4:170, 5:9, 5:18, 5:66, 5:106
Apache	1:13, 1:31, 1:58, 1:256, 2:42, 3:65, 3:114, 4:110, 4:124, 5:64, 6:41
API Hooking	5:45, 5:48
APNIC	2:20
APOP-MD5	4:26
APPEVENT.EVTX	5:114
application log	1:53, 1:58, 1:125, 4:111, 4:122
APPLICATION.LOG	5:114
archive.org	2:39
ARIN	1:65, 1:104, 1:143, 1:167, 2:10, 2:12, 2:20, 2:30, 2:70-71, 2:137, 3:129, 4:27, 4:49, 4:57, 4:101, 4:176, 4:184, 5:16, 5:37, 5:60, 5:105, 5:171, 6:23, 6:41
Armitage	3:113
ARP-cache-poisoning	4:24
Arpspoof	3:51-52
Aruba	2:75
ASCII	1:169, 3:67, 3:105, 3:107, 3:153, 4:114, 4:133, 4:147, 5:85-86, 5:90, 5:127-130
Assessment Questions	1:95-96
Assigning Handlers	1:50
assumptions	1:115, 2:110, 4:24
at	1:32, 1:35, 1:63-66, 1:68-74, 1:80, 1:82-88,

Beacons	143 1:146, 1:149-151, 1:153-154, 2:60, 5:136, 5:164
Bell-LaPadula	4:123
Bettercap	3:51, 3:53, 3:58-59
BeyondTrust	2:119
bg	1:227
Bind shell	3:115
Binders	5:18
Bing	1:47, 1:107, 1:208, 1:261, 2:11-12, 2:29, 2:35, 2:43, 2:83, 2:132, 3:25, 3:97, 4:27, 4:49, 4:60, 4:64, 4:108, 4:123, 4:166, 5:167-168, 5:180, 6:12
BlackShades	5:9
Blaster	4:55
Bloodhound	2:143
Blue Coat	1:172
Blue Pill	5:58
bonk	4:154, 4:161
Border Gateway Protocol (BGP)	3:2, 3:8-9, 5:192
Border Gateway Protocol (BGP4)	3:8, 5:192
Bot Distribution	4:67
botherder	4:66, 4:69
botnet	1:109, 1:134, 2:14, 4:55, 4:60, 4:66, 4:68- 70, 4:161, 4:163, 4:165, 4:167
Bridged	1:8, 1:202, 1:204, 6:3-4, 6:8-9
Bridged network	1:8, 1:202, 6:4, 6:8-9
Bro	1:147-148, 1:153-158
Browser Exploit Against SSL/TLS (BEAST)	3:66
Browser Exploitation Framework (BeEF)	3:51, 4:76, 4:79, 4:81-83, 4:120
btmpt	5:89-90
buffer overflow	1:256, 2:137, 3:1-2, 3:46, 3:95-109, 3:112, 3:121-127, 3:129, 3:131-134, 4:57, 4:74, 4:147, 4:161, 5:10, 5:41, 5:167, 6:35
Bugtraq	1:15, 1:95, 5:197-198
Building a Team	1:30
Burp	4:91, 4:93-95, 4:104, 4:146
Burp Suite	4:104, 4:146
Business Continuity Planning (BCP)	1:30

cone of silence	5:58, 5:61
Conficker	4:55, 4:57-58
Containment	1:2, 1:14, 1:17, 1:95-96, 1:99-115, 1:117, 1:183, 2:39, 2:58, 2:75, 2:86, 3:86, 3:129, 4:52, 4:74, 4:101, 4:111, 4:126, 4:151, 4:171, 5:69, 5:145, 5:156
Containment Checklist	1:14
Coverity	3:127
Covert Channel	5:4, 5:120, 5:124, 5:127-128, 5:130, 5:132, 5:134, 5:136-145, 6:39
Covert_TCP	5:4, 5:127-131, 5:134
CoWPAtty	2:65
CpuHog	4:154
cracking modes	4:38
CRAM-MD5	4:26
CreateRemoteThread	5:45
crontab	1:253, 5:44
Cross-Site Scripting (XSS)	4:3, 4:76, 4:110, 4:113-126, 4:128-141, 4:147, 4:150, 5:168, 6:59
CUDA	4:10
cyber warfare	5:192
CyberCPR	1:105
CyberSponse	1:104

D

Dameware	5:9
Data Execution Prevention (DEP)	3:123
Data Loss Prevention (DLP)	1:192, 2:43, 4:111, 5:133
dd	1:41, 1:110, 1:178, 5:22, 5:25-27, 5:34
DefCon	1:15, 5:196
Denial of Service (DoS)	1:10, 1:27, 1:95, 1:102, 1:109, 1:185, 2:7, 2:11, 2:14, 2:50, 2:75, 2:122, 3:112, 4:3, 4:65, 4:73, 4:153-154, 4:161, 4:163-171, 4:173, 4:175-176, 4:181, 4:183, 6:17
df	1:4-5, 1:13-14, 1:23, 1:62-63, 1:105, 1:257, 2:2, 2:40-42, 2:96-97, 2:129, 3:53, 3:131, 4:20, 4:51, 4:171, 6:49
DHCP leases	2:68
Dictionary attack	2:64, 4:10-11, 4:27, 6:49
Dig	1:4-5, 1:43, 1:92, 1:101, 1:197, 2:9, 2:24, 2:26, 2:43, 2:54, 2:57, 2:153, 3:65, 3:72,

Error messages	1:256, 3:9, 3:12, 3:18, 3:85, 4:91, 4:96, 4:104, 4:106
Espionage	1:3, 1:160-164, 5:156
Ethereal	3:46
Ettercap	2:68, 3:51, 3:53, 3:58-59, 3:64
Event Definition	1:12
Event Viewer	1:74, 1:90, 3:164, 3:166, 5:16, 5:114
eventquery.vbs	1:74, 1:191
eventvwr.msc	1:74, 1:90, 3:164, 3:166, 3:173
Evilgrade	2:12, 3:148
Excel	1:20, 1:28, 1:136, 1:139-140, 2:6, 2:10, 2:40-41, 2:43, 2:80, 2:98, 3:133, 4:62, 4:137, 4:147, 5:18, 5:22, 5:64, 5:136, 5:149
Exe32pack	5:19
Exploit Database	2:35
explorer.dll	5:48
explorer.exe	3:168, 5:39, 5:46, 5:48, 5:109-110
Extension Mechanisms for DNS (EDNS)	4:157, 4:159
External Hacking	1:102
External mode	4:38
Extortion	2:9, 2:11
EyeWitness	2:98

F

Facebook	2:30
Fast Flux	4:69-71
fg	1:227-228
Fgdump	1:57, 4:29, 4:31, 4:35, 4:49, 6:16
fgets	3:96, 3:103
Fiddler	4:146
filetype:	2:40
FIN	2:90, 2:93, 2:95
Firebug	4:144
Firefox	1:149, 2:122, 2:126, 3:61, 3:114, 4:114, 4:125, 4:132, 4:144, 5:139
First in First Out (FIFO)	3:23-25, 3:35
Flame	3:146, 4:55
Flash	1:12, 1:47, 2:43, 3:114, 3:133, 4:60-61, 4:77, 4:82-85, 4:87, 4:125, 5:34, 5:36-37, 5:39, 5:82, 5:149, 5:190
FlashUpdate.exe	4:77, 4:83, 5:34, 5:36-37, 5:39

Half-Open	2:93, 4:173, 4:175, 4:179-180, 4:185-186, 4:188
Hashcat	3:91, 4:10
hashdump	3:152, 3:162, 4:29, 4:49, 6:51
HBGary	5:22
heartbeat	1:151, 3:65
Heartbleed	3:65
Hex	1:169, 2:153-154, 3:72, 3:74, 3:105, 4:114, 4:126, 4:133, 4:147, 5:140, 5:142
Hidden Form Elements	4:143-144, 5:136
Hiding Files	5:3, 5:53, 5:82, 5:104-106
High Orbit Ion Cannon (HOIC)	4:169
High Technology Crime Investigation Association (HTCIA)	1:25
Hijacking	2:68, 3:1-2, 3:4, 3:8-9, 3:64, 3:79-86, 3:88, 3:146
HKEY_CURRENT_USER (HKCU)	1:69, 1:85
Honeynet Project	1:22, 5:196
Hop Limit	2:81, 2:83
Host Info (HINFO)	2:25
Host-only network	1:202, 3:154, 4:176
HP SPI Dynamics	4:146
HTTP Strict Transport Security (HSTS)	3:68
Human Interface Devices (HID)	2:74, 3:5
Human Resources	1:30, 1:166, 1:170, 3:77
Hybrid attacks	4:13, 4:27
Hydan	5:4, 5:148, 5:150-153, 6:58
Hydra	4:8, 5:164-168
HyperText Transfer Protocol (HTTP)	1:155, 1:169, 1:177, 2:42, 2:67-68, 2:70, 2:90, 3:21, 3:64, 3:67-68, 3:70, 3:80, 3:128, 3:157, 4:8, 4:66, 4:68, 4:71, 4:73, 4:98, 4:119, 4:123, 4:125, 4:133-134, 4:143-147, 4:154, 4:167-169, 5:4, 5:122, 5:124-125, 5:127, 5:134, 5:136-138, 5:141, 5:171
I	
ICMP	1:233, 2:82-83, 2:85, 2:93, 2:96, 4:99, 5:4, 5:120, 5:124-125, 5:127, 5:131, 5:134, 6:3, 6:11
Identification	1:2, 1:14, 1:17, 1:49-61, 1:94, 1:97, 1:100-

Internet Relay Chat (IRC)
 Internet Storm Center (ISC)
 Internetwork Operating System (IOS)
 InterNIC
 intitle:
 Intrusion Detection System (IDS)

Intrusion Prevention System (IPS)

inurl:
 Invisible Secrets
 IP Identification field
 IP packet header
 ipconfig
 iptables

ISP Coordination
 iTunes

J

Java

Jikto
 John The Ripper

john.pot
 jolt

3:76-77, 3:84, 3:89, 3:111, 3:113, 3:153,
 3:157, 3:171, 4:69-71, 4:73, 4:77, 4:82,
 4:96, 4:99, 4:111, 4:119-120, 4:134, 4:181-
 182, 4:186, 5:25, 5:30, 5:34, 5:120, 5:125,
 5:127-131, 5:134, 5:162, 5:164, 5:166,
 5:180-181, 5:184, 5:194, 6:3, 6:6-9, 6:13-
 15, 6:21, 6:27, 6:29, 6:32-33, 6:51
 4:66, 4:68-69, 4:164, 4:168
 1:15, 1:52, 2:6, 5:195
 4:26, 5:192
 2:19, 2:53, 5:162, 6:25
 2:37, 2:42
 1:13, 1:20, 1:53, 1:125, 1:146, 1:158, 1:162,
 1:164, 1:168, 1:170, 1:177, 1:187, 1:193, 2:3,
 2:75, 2:85, 2:100, 2:114-116, 2:124, 2:141,
 2:148, 3:117-118, 3:129, 4:62, 4:126,
 4:170-171, 5:132-134
 1:20, 1:44, 1:53, 1:114, 1:125, 1:146, 1:158,
 1:162, 1:187, 1:193, 2:3, 2:58, 2:114-116,
 2:141, 3:117-118, 3:121, 3:129, 4:52, 4:74,
 4:170, 5:68, 5:132-133
 2:37, 2:42
 5:148
 5:128
 2:81
 1:86, 3:85, 3:163
 1:203, 2:151, 3:28, 3:30, 3:154, 4:176, 6:8,
 6:51
 1:109
 2:12, 3:114, 3:133

2:12, 3:54, 3:66, 3:114, 3:116, 3:125, 3:127,
 4:76, 4:87, 4:89, 4:104, 4:113-118, 4:120,
 4:124-125, 4:146, 4:168-169, 5:12, 5:154,
 5:190, 6:59
 4:119
 3:91-92, 4:2, 4:10, 4:13, 4:27, 4:35-41,
 4:43-47, 4:141, 6:16, 6:52
 4:39, 6:52
 4:161

Lightweight Extensible Authentication Protocol (LEAP)	2:60, 2:64
Link-Local Multicast Name Resolution (LLMNR)	3:6, 3:50-51, 3:79-80, 3:84, 3:88-89
link:	2:29, 2:37
LinkedIn	2:12, 2:30
LMHash	4:18, 4:32
lockout	2:139-140, 2:161, 4:6-7, 4:94, 4:96
log2timeline	1:43
long-tail analysis	1:140
Low Orbit Ion Cannon (LOIC)	4:168-169
ls	1:148, 1:155, 1:208, 1:212, 1:215, 1:217, 1:221-224, 1:227-229, 1:235, 1:239, 1:242, 1:250, 1:253, 1:261-262, 2:25, 2:144, 3:33, 3:117, 3:171, 4:77, 5:44, 5:50, 5:74, 5:82, 6:29, 6:47, 6:58
lsuf	1:56, 1:234, 1:247, 1:250, 1:252, 1:260, 1:262-263, 2:103, 5:76, 5:95, 5:100
lusrmgr.msc	1:71, 1:88

M

Mac OS X	2:12, 2:46, 2:122, 3:11, 3:46, 3:116, 5:11, 5:67
Macro_safe.py	3:142
Mail eXchanger (MX)	2:25, 5:177
Maltego	2:2, 2:46-48
Malware	1:30, 1:69-70, 1:102, 1:119, 1:126, 1:134-136, 1:144, 1:146, 1:196-198, 2:12, 2:14, 2:43, 3:4-5, 3:51, 3:59, 3:65, 3:67, 3:70, 3:77, 3:137-139, 3:146-150, 4:55, 4:58-59, 4:76-77, 4:83, 4:87, 4:123, 5:2, 5:7, 5:18, 5:20, 5:37, 5:53, 5:55, 5:68, 5:99, 5:102, 5:132, 5:136
Malware Domain List (MDL)	1:134
Malware layers	5:7
man	1:208, 1:212-213, 1:222, 1:229, 1:235, 1:239, 1:242-243, 2:145, 3:63, 5:83, 5:96, 5:142-143
Man-in-the-Middle Framework (MitMf)	3:51, 3:53-54, 3:68, 3:83
Management Support	1:29
Masscan	2:97

nanobotninja	1:154, 1:156-157
nbtstat	1:65
nc	1:81, 1:260, 1:262-263, 2:111-112, 3:14, 3:16-17, 3:19-23, 3:25, 3:29-36, 3:38, 3:42, 4:133, 4:188, 5:18, 5:75, 5:77, 5:93- 94, 5:98-99, 5:110, 6:55
ncat	1:41, 3:11, 3:125, 4:11, 4:13
Nessus	1:121, 2:3, 2:119-123, 2:126-127, 2:132, 2:135, 3:17, 6:16, 6:31
Nessus Attack Scripting Language (NASL)	2:123
nestea	4:161
net localgroup	1:71, 1:88, 3:163, 6:53
net session	1:65, 2:146, 2:150, 2:156-157, 2:162, 4:52
net sessions	4:52
net use	1:65, 1:71, 1:126, 2:137-140, 2:146, 2:150, 2:156, 2:162, 3:163, 4:50, 6:36, 6:53
net view	1:65, 2:139-140, 2:152
NetBIOS Name Service (NBT-NS)	2:148, 3:50
Netcat	1:41, 1:57, 1:81, 1:260, 1:262-263, 2:111- 112, 2:135, 3:2, 3:11-26, 3:28-43, 3:115, 4:73, 4:133, 4:164, 4:179, 4:181, 4:185, 4:187-188, 5:6, 5:12, 5:18, 5:22, 5:35, 5:49, 5:73, 5:75, 5:77, 5:98-102, 5:110, 6:16, 6:37-38, 6:55
NetMon	3:134
NetNanny	1:172
netsh	1:66, 1:92, 1:203, 2:73-74, 2:151, 3:28, 3:30, 4:176, 6:8
netsh advfirewall	1:66, 1:92, 1:203, 2:151, 3:28, 3:30, 4:176, 6:8
netsh firewall	1:66, 1:203, 2:151, 3:28, 3:30, 4:176, 6:8
netsh wlan	2:73-74
Netsky	4:55
netstat	1:55-56, 1:66, 1:81, 1:208, 1:212, 1:222, 1:229, 1:234-235, 1:239, 1:241, 1:252, 2:101, 2:103, 3:164-165, 3:169, 3:173, 4:179-180, 4:182, 4:185-188, 5:23, 5:25, 5:33, 5:44, 5:47-48, 5:63, 5:76, 5:96, 5:101
NetStumbler	2:61, 4:24
Network Address Translation (NAT)	1:136, 1:202, 3:11, 6:4
Network Management	1:30, 2:94, 4:74, 5:134
Network Operations Center (NOC)	1:8, 2:31
newtear	4:161

P

packet fragmentation	5:128, 5:162
Packet Storm	5:195
passwd	1:126, 1:208, 1:210, 1:212-213, 1:220, 1:222, 1:229, 1:235, 1:239, 1:254, 1:264- 265, 2:132, 4:20, 4:35-37, 4:41, 4:43-45, 5:43, 5:96, 5:101-102, 5:115, 6:15, 6:52
password cracking	2:73, 4:2, 4:5-14, 4:22-24, 4:26-27, 4:31, 4:33, 4:40, 4:47, 4:49, 4:139, 5:170, 5:189, 6:16
password guessing	2:56, 3:157, 3:173, 4:6-8, 4:55, 6:16, 6:35
Password List (PWL)	1:249, 4:26
password resets	4:7
password spraying	2:159, 4:7
PEBundle	5:19
PECompact	5:19
PeepNtom	2:98
permission	1:35, 1:190-191, 1:193, 1:209, 1:215, 1:249, 1:251, 1:261, 2:7-8, 2:48, 2:66, 2:78, 2:143, 3:20, 3:42, 3:106, 4:36, 4:72, 4:110, 5:52, 5:57, 5:86, 5:100, 5:102, 5:114, 5:144, 6:3, 6:11, 6:28, 6:45-46
Personally Identifiable Information (PII)	1:23, 4:111, 5:159
phpBB	2:42
Picasa	2:31
ping	1:8, 1:106, 1:203, 1:208, 1:212, 1:222, 1:229, 1:233, 1:235, 1:239, 2:50, 2:82, 2:84-86, 2:93-94, 2:151, 3:154, 4:99-100, 4:176, 5:124-125, 5:164, 6:3, 6:8, 6:11
Ping of Death	4:154
Pivot	1:144, 2:36, 2:118, 3:25-26, 3:29, 3:117- 118, 4:98, 5:24, 5:29, 5:33, 5:39, 5:101
PIX	4:24, 4:26, 5:154
Pluggable Authentication Modules (PAM)	4:40-41
podcasts	5:199
Poison Ivy	5:9, 5:14-15, 5:18
polymorphic	3:119, 4:56, 4:62-65, 5:151, 5:194
POP3	2:70, 4:8
port 53	2:25, 2:27, 2:93, 3:16, 5:162
port sentry	1:53
portmapper	2:93
positive skew analysis	1:140

Q

Quality Assurance (QA)	5:45
Qualys	1:121, 2:119
Quick UDP Internet Connection (QUICK)	5:132

R

RADIUS Shared Secrets	4:26
Rainbow Tables	4:22, 4:27, 4:139, 5:173
Ramen	4:57-58
Ransomware	2:10, 4:55
Rapid7	1:121, 2:54, 2:119, 5:117
Real intelligence Threat Analytics (RITA)	1:147-148, 1:150, 1:154-155, 1:158, 3:128, 5:68
Real Time Incident Response (RTIR)	1:104
Recon-ng	2:31, 2:43
Reconnaissance	1:56, 1:95, 2:2, 2:16, 2:19, 2:22, 2:27, 2:35, 2:37, 2:40, 2:44, 2:46, 2:48, 5:162, 6:2, 6:22-23, 6:28, 6:30, 6:34, 6:37, 6:39, 6:44
Recovery	1:2, 1:10, 1:14, 1:17, 1:30, 1:32, 1:42, 1:103, 1:109, 1:114, 1:118, 1:123-126, 1:128, 2:58, 2:75, 3:129, 4:52, 4:74, 4:101, 4:111, 4:126, 4:151, 5:69, 5:145
reg query	1:69, 1:85, 1:192
regedit	1:69, 1:85, 4:50
Registration Authority (RA)	3:65
regsvr32	3:150
Rekal	1:39, 1:110, 5:22-27, 5:29, 5:31-35
Rekall	1:39, 5:22-27, 5:29, 5:31-35
rel	1:208, 1:212, 1:222, 1:229, 1:235, 1:239
related:	2:37
Relay	1:167, 3:11, 3:15, 3:22-26, 3:28, 3:35, 3:40-43, 3:117, 4:66, 4:73, 4:164, 5:39
Remote Desktop Protocol (RDP)	2:42, 2:98, 2:115, 4:8, 4:55
Remote Procedure Call (RPC)	1:83, 1:256, 2:93, 2:109, 2:148, 3:17, 3:123, 3:131, 5:37
Remote Procedure Calls (RPC)	1:83, 1:256, 2:93, 2:109, 2:148, 3:17, 3:123, 3:131, 5:37
remote system logging	5:118
Remux	2:99
Reserved	1:1, 2:1, 2:88, 3:1, 4:1, 4:19, 4:37, 5:1,

SANS Investigative Forensics Toolkit (SIFT)	1:41-43
Santy	2:42
Sasser	4:55, 4:57-59
sc query	1:68, 1:83, 2:102
scanf	3:103
Scareware	5:16
Scheduled Tasks	1:62, 1:73, 1:86-87, 1:253
schtasks	1:73, 1:87
SearchDiggity	2:43
SECEVENT.EVTX	5:114
secpol.msc	1:89, 5:45, 6:33
Secure SHell (SSH)	1:119, 1:175, 1:187, 2:115, 3:57, 3:60, 3:63, 3:65, 3:84-85, 3:128, 4:8, 5:10, 5:39, 5:43, 5:120, 5:125, 5:174, 5:190, 6:54
SecurID	4:24
Securing the Human	1:20
Security Event Management (SEM)	1:104
Security Information Management (SIM)	1:104, 2:94
Security Onion	5:68
SECURITY.LOG	5:114
Separation of duties	5:178
Sequence Number field	5:128-130
Server Message Block (SMB)	1:65-66, 2:3, 2:89, 2:109, 2:137-140, 2:142-148, 2:150, 2:156-162, 3:80, 3:92, 3:155, 3:173, 4:8, 4:30, 4:49-52, 4:57, 4:101, 5:22, 5:37, 5:39, 6:36, 6:51
services.msc	1:68, 1:83, 2:102
session	1:65, 1:206-207, 1:225, 2:3, 2:7, 2:67-68, 2:88, 2:94, 2:112, 2:114, 2:137-140, 2:142-148, 2:150, 2:156-157, 2:162, 3:1, 3:6, 3:9, 3:21, 3:26, 3:54, 3:60, 3:64, 3:67, 3:83-86, 3:92, 3:118, 3:158-159, 4:26, 4:52, 4:68, 4:86, 4:111, 4:117-118, 4:128-129, 4:132-133, 4:135, 4:139, 4:141, 4:143, 4:145, 4:148-149, 5:10, 5:89, 5:101, 5:129, 5:136-137, 6:22, 6:31, 6:36, 6:51-53
sessionID	4:143, 4:149
sessions -l	3:159
set LHOST	3:157, 3:172, 4:78, 6:51
set RHOST	3:157, 6:51
set SMBPass	3:157, 6:51
set SMBUser	3:157, 6:51

strcpy	3:96, 3:103, 3:107
Stream Control Transmission Protocol (SCTP)	5:132
Street View	2:36
strncpy	3:96, 3:103, 3:107
Structured Query Language (SQL)	1:96, 3:80, 4:3, 4:8, 4:26, 4:55, 4:103-111, 4:124, 4:128-130, 4:136-137, 4:141, 4:147, 4:150, 5:64, 5:168-172, 5:174
Stuxnet	3:65, 3:146, 4:55, 4:57-59, 4:64, 5:55
su	1:7, 1:208, 1:211-212, 1:222, 1:229, 1:231, 1:235, 1:239, 1:244, 2:106-107, 2:111, 3:38, 3:89, 3:153, 3:171-172, 4:43, 4:77-79, 4:130, 4:180, 5:43, 5:76, 5:79, 5:138, 5:141, 5:184, 6:7, 6:51, 6:56
Sub7	5:9
Subterfuge	3:64
SubVert	1:160, 2:12, 4:98, 5:55, 5:58
SucKIT	5:56, 5:65
sumfuq	5:50
Supervisory Control And Data Acquisition (SCADA)	3:114, 4:58, 4:64
Survey	1:14, 1:101, 1:182-183, 3:5
Suspicious Activity Report (SAR)	1:23
Suterasu	5:71-74
Switch User ID (SUID)	1:249, 1:261, 3:106
Symantec	3:29, 3:133, 4:119
SYN	1:54, 2:25, 2:82, 2:90, 2:93-95, 2:97, 3:36, 3:38, 4:73, 4:154, 4:165, 4:167, 4:171, 4:175, 4:179-180, 4:185-186, 5:129-130, 2:25, 2:90, 2:93-94, 4:165, 4:185
SYN-ACK	4:161
syndrop	5:51, 5:53, 5:55
SYS_execve	5:114
SYSEVENT.EVTX	1:69, 1:77, 1:126, 1:141, 2:137, 5:66, 5:106
Sysinternals	3:162, 4:31
SYSKEY	5:44, 5:85, 5:117-118
syslog	5:114
SYSTEM.LOG	2:104, 2:126, 2:135, 4:185-186, 4:188
systemctl	
T	
Tamper Data	4:144

Trend Micro	3:157, 3:164-165, 3:169, 4:68, 4:73, 4:101, 4:119, 4:165, 4:167-168, 4:173, 4:175-176, 4:179, 4:181-182, 4:185-186, 4:188, 5:10, 5:12, 5:23, 5:37, 5:43, 5:47, 5:49, 5:52, 5:75, 5:100-101, 5:124-125, 5:127-132, 5:134, 5:162, 5:165, 5:168, 6:6
Tribe Flood Network (TFN)	3:133
Tribe Flood Network 2000 (TFN2K)	4:163
Tripwire	4:163
Trivial File Transfer Protocol (TFTP)	1:193, 1:258, 5:7, 5:50, 5:67
Trojan Horse	2:129, 2:132
	1:224, 2:12, 4:64, 5:2, 5:6, 5:9-16, 5:18, 5:22, 5:41, 5:50, 5:131, 6:38
Tunneling	5:120, 5:127
Twitter	2:30-31, 3:147, 4:68, 4:168, 5:171-186

U

UDP 137	2:148
UDP 138	2:148
UDP 53	2:89
UDP Header	2:89, 2:92, 5:128
uname	5:93, 5:98
Unauthorized use	1:11, 1:21, 1:166-172, 1:175
Unicode	1:169, 4:18, 4:114, 4:126
Union Representation	1:30
Universal Plug aNd Play (UPnP)	4:59
unlink	1:250, 1:262
uptime	1:257
URG	1:126, 2:13, 2:48, 2:86, 2:90-91, 2:95, 4:18, 4:32, 4:62, 4:67, 4:156, 5:128, 5:166
US-CERT	5:198
UseNet	2:29
User Account Control (UAC)	1:81
User Agents	1:150
User Behavioral Analytics	2:116, 5:117
User Datagram Protocol (UDP)	1:62, 1:66, 1:77, 1:81, 1:234, 1:241, 1:247, 1:252, 2:27, 2:88-89, 2:92-93, 2:95, 2:101, 2:103, 2:107, 2:137, 2:148, 3:11, 3:13-14, 3:16-19, 3:31, 3:33-34, 3:47, 3:111, 3:165, 4:73, 4:156-157, 4:159, 4:168, 4:179, 5:23, 5:43, 5:47, 5:52, 5:75-76, 5:124, 5:128,